SIXTH ANNUAL ECC
UNDERGRADUATE
MATHEMATICS COMPETITION

March 29, 2003

Time control: Three hours

To the team members: We hope you find these problems to be fun as well as challenging. If you do not succeed in working all ten of them in the allotted time, you are likely to be in good company. Each problem counts 10 points. Partial credit will be given for significant progress or for significant partial solutions, but a thorough job on a few of them will probably be better than some exploratory work on many.

NO BOOKS, NOTES, CALCULATORS, COMPUTERS OR NON-TEAM-MEMBERS may be consulted.

Each team may submit one solution to each problem. Think of your solution as an essay; a logical argument which makes clear why your answer to the question is correct, or why the assertion whose proof is called for in the problem is true.

PLEASE BEGIN EACH PROBLEM ON A NEW SHEET OF PAPER. Team identification and problem number should be clearly given at the top of each sheet of paper submitted.

1. Right triangles.

A triangle T has sides

$$14 \cdot 10^{n+1}, \quad 245 \cdot 10^{2n+1} - 2, \quad \text{and} \quad 245 \cdot 10^{2n+1} + 2,$$

where n is a positive integer. Prove that T is a right triangle.

2. Probability of solving.

Three members of a team work independently on a problem. If each one has probability $\frac{1}{3}$ of solving the problem, what is the probability that at least one of the three solves it?

3. A differential equation.

Suppose that $f'(x) = xf(x)$ for all x, where f is a twice differentiable function, and that $f(-3) = 2$. Find $f''(-3)$, and justify your answer.

4. Maclaurin series.

Let $\sum_{k=0}^{\infty} a_k x^k$ be the Maclaurin series for

$$f(x) = \int_0^x \frac{1 - \cos t}{t^2} dt.$$

Find a_{11}, and justify your answer.

5. Solve for $\frac{x}{y}$.

If $2\ln(3x - y) = \ln(2x) + \ln(2y)$, what are the possible values of $\frac{x}{y}$?

6. An inequality.

For what x is $x < 2x + 1 < x^2$? Give your answer in the form of an interval or union of (possibly infinite) intervals.

7. Find the limit.

Let

$$f(x) = \left(\frac{3^x + 5^x}{2}\right)^{\frac{1}{x}}.$$

Find the following limits, and justify your answers:

(a) $\lim_{x\to\infty} f(x)$;

(b) $\lim_{x\to 0} f(x)$.

8. The BEDCAF dictionary.

The BEDCAF people have an alphabet consisting of the letters A, B, C, D, E, F, in that order. The words in their language consist of precisely all the permutations of these six letters; i.e., all six-letter strings which use each letter just once. The words in the BEDCAF dictionary are, of course, in alphabetical order, and begin ABCDEF, ABCDFE, ... What is the 525th word in the BEDCAF dictionary? Justify your answer.

9. Some sum.

For integers $n > 0$ define $f(n)$ to be the integer nearest to $\sqrt[4]{n}$. Evaluate

$$\sum_{k=1}^{2003} \frac{1}{f(k)}.$$

10. How many terms?.

When $(x + y + z)^2$ is expanded and like terms are collected, one gets

$$x^2 + y^2 + z^2 + 2xy + 2xz + 2yz,$$

for a total of 6 terms. Determine, with proof, the number of terms obtained when

$$(x + y + z)^{2003}$$

is expanded and like terms are collected.